

1. Use `systemctl` to view the status of all the system services.

```
systemctl list-unit-files --type=service
```

2. Change the default run level back to `multi-user.target` and reboot.

```
systemctl isolate multi-user.target
```

3. Send mail to the root user.

```
mail -s "Best regards, Ahmed Ehab" root
```

4. Verify that you have received this mail.

```
Su -  
mail
```

5. Use `systemctl` utility to stop `postfix` service

```
systemctl stop postfix
```

6. Send mail again to the root user.

```
mail -s "Another mail" root
```

7. Verify that you have received this mail.

```
Su -  
mail  
There is no new mails
```

8. Use `systemctl` utility to start `postfix` service

```
systemctl start postfix
```

9. Verify that you have received this mail.

```
Su -  
mail  
Found the mail i sent in Q6
```

10. Edit in the GRUB2 configuration file and change the timeout variable equal 20 seconds.

```
sudo vi /etc/default/grub  
GRUB_TIMEOUT=30  
sudo update-grub
```

11. Edit in the GRUB2 configuration file and change your default operating system

```
sudo vi /etc/default/grub
GRUB_DEFAULT=saved
GRUB_SAVEDEFAULT=true
sudo update-grub
```

12. You want to know some information about the status of the system every ten minutes today between the hours of 8:00 AM and 5:00 PM. to help investigate some performance issues you have been having. You suspect it might be memory related and want to keep an eye on those resources.

```
crontab -e
*/10 8-17 * * * /usr/bin/free -m
```

13. Use mail as the root user to check for e-mail from the cron jobs you have scheduled.

```
Su -
mail
```

14. How could you send the output from these cron jobs to another e-mail address (the manager user)?

```
crontab -e
MAILTO=manager
```

15. Use mail as the manager user to check for e-mail from the cron jobs you have scheduled.

```
su manager
mail
```

16. Grant your colleagues a collective directory called /opt/research, where they can store generated research results. Only members of group profs and grads should be able to create new files in the directory, and new file should have the following properties:

- the directory should be owned by root
- new files should be group owned by group grads
- group profs should automatically have read/write access to new files
- group interns should automatically have read only access to new files
- other users should not be able to access the directory and its contents at all.

```
mkdir -p /opt/research
chown root:grads /opt/research
chmod 2770 /opt/research
setfacl -m g:profs:rwX /opt/research
setfacl -d -m g:profs:rw /opt/research
setfacl -d -m g:interns:r /opt/research
chmod 770 /opt/research
```

17. Display your MAC address by 2 different ways.

```
ip link show
nmcli device show
```

18. Display the network settings of all active interfaces.

```
nmcli device status
```

19. Display the network setting of all interfaces both active inactive.

```
nmcli connection show
```

20. Bring your interface down.

```
nmcli device disconnect <interface>
```

21. Configure your network card to have static IP.

```
nmcli con mod <connection_name> ipv4.addresses 192.168.1.10/24 ipv4.gateway 192.168.1.1 ipv4.method manual
```

22. Bring your interface up.

```
nmcli con up <connection_name>
```

23. Verify your network setting using ifconfig command

```
nmcli device show <interface>
```

24. Configure your network card to have dynamic IP using network manager command.

```
nmcli con mod <connection_name> ipv4.method auto  
nmcli con up <connection_name>
```

25. Check using ifconfig then check its configuration file.

```
cat /etc/NetworkManager/system-connections/<connection_name>.nmconnection
```

26. Reconfigure your network card using system-config-network utility to have static IP.

```
nmcli con edit <connection_name>
```

27. Configure your network card to have 3 IPs and check that they are all working using ifconfig command.

```
nmcli con mod <connection_name> +ipv4.addresses 192.168.1.11/24  
nmcli con mod <connection_name> +ipv4.addresses 192.168.1.12/24  
nmcli con up <connection_name>
```

28. Change your host name in your global network file.

```
nmcli general hostname <new-hostname>
```

Your boss thinks it's a great idea to have one central logging server. Satisfy his requirements ☺

Hint:

Set up rsyslogd on the "logging server" machine to accept logging messages from other machines.

On the your "workstation", set up rsyslogd to send messages to the "logging server"

Test the new setup by using the logger command on the "workstation" to generate a log message

Does the message appear in the "logging server's" /var/log/messages file?

Why does this message also appear in the "workstation's" /var/log/messages file?

29. How could you have the message only appear in the "logging server's" files?

```
sudo vi /etc/rsyslog.conf
module(load="imudp")
input(type="imudp" port="514")
*. * @<SERVER_IP_ADDRESS>:514
& stop
systemctl restart rsyslog
firewall-cmd --add-port=514/udp --permanent && firewall-cmd --reload
```

```
logger "Hello Boss, the central logger is working"
tail /var/log/messages
```